Notice, question, predict

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Learning target: Use observations to make a prediction that can be tested.

Suggested first pass: 1, 2, 3, 9. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Read a scale (retrieve)
Marks 5 cm apart; rightward arrow from 2 to 7 cm.

02. Explain a worked example (worked)
Four equally spaced points; the 11 m point is the prediction. A new measurement could disagree.

03. Notice a pattern (model)
Positions increase by 2 m each second; the cart moves right with equal displacements in equal times. Accept observations tied to the displayed values.

04. Make a prediction (practice)
7 m: another +2 m from 5 m. Assumes the same velocity continues.

05. Evidence or a claim? (reason)
E, C, E. Approaching alone does not establish impact.

06. Test the model (reason)
3 m if the -2 m each second pattern continues. Check measurement quality and whether motion changed.

07. Ask a useful question (reason)
Examples: Where will it be in a month? Need repeated positions, velocity direction and time. How large is it? Need observations of size. Accept a measurable question and matching evidence.

08. Choose useful evidence (mixed)
The measured times with the track distance allow calculation and repetition. The phrase alone gives no measurable comparison.

09. Exit check (exit)
12 m, assuming +3 m each second continues. A later position inconsistent with that prediction, beyond measurement variation, challenges the model.

10. STRETCH: Apply it another way (stretch)
Many answers: continuing gives 14 m; stopping at 2 s gives 8 m. Observe later positions and times.

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01. Read a scale (retrieve)
Marks 5 cm apart; rightward arrow from 2 to 7 cm.

02. Explain a worked example (worked)
Four equally spaced points; the 11 m point is the prediction. A new measurement could disagree.

03. Extract the evidence (model)
Approach and uncertainty are stated. Impact, a precise arrival date, or a damage estimate is not established.

04. Fit a simple model (practice)
$x = 1 + 3\,t$ in meters, with t in seconds; $x(5)=16$ m. Assumes constant velocity beyond the observed interval.

05. Challenge the prediction (reason)
Prediction 16 m lies outside 13.8-14.2 m. Motion may have changed or earlier measurements/model assumptions may be wrong. Do not infer a specific cause without evidence.

06. Design the observation (reason)
Positions in a stated frame at two known times estimate velocity including direction. Other justified observations accepted; must explain how they improve the prediction.

07. Compare claims (reason)
Need positions of both bodies at the same time, gravitational/orbital changes, 3 D state and uncertainty. A simplified model alone cannot establish real impact.

08. Choose useful evidence (mixed)
The measured times with the track distance allow calculation and repetition. The phrase alone gives no measurable comparison.

09. Exit check (exit)
$v=-2$ m/s; $x=12-2(7)=-2$ m. Assumes constant velocity. Measure x at a stated later time and compare with the model including uncertainty.

10. STRETCH: Apply it another way (stretch)
One rule $x=2+3\,t$ throughout; another follows that rule through 2 s then stays at 8 m. At 4 s: 14 m versus 8 m; future behavior is unmeasured.